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En-ConvNet: A novel approach for glaucoma detection from color fundus images using ensemble of deep convolutional neural networks

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Abstract

Glaucomatous optic neuropathy is the preeminent cause of incurable vision impairment and blindness across the world. Manual interpretation of the pathological structures in fundus images is time-consuming and requires the expertise of a competent specialist. With the development of deep learning approaches, automated glaucoma diagnosis is easy and effective for larger screening. Convolutional neural networks, in particular, have emerged as a promising choice for glaucoma detection from fundus images due to their remarkable success in image classification. Transferring the optimized weights from a pre-trained model expedites and simplifies the training process of deep neural network. In this paper, a deep ensemble model using the stacking ensemble learning technique is developed to attain the optimum performance for the classification of glaucomatous and normal images. Thirteen pre-trained models such as Alexnet, Googlenet, VGG-16, VGG-19, Squeezenet, Resnet-18, Resnet-50, Resnet-101, Efficientnet-b0, Mobilenet-v2, Densenet-201, Inception-v3, and Xception are implemented. Their performance is compared in 65 different configurations, comprising 13 CNN architectures and five various classification approaches. A two-stage ensemble selection technique is proposed to select the optimal configurations. Selected configurations are pooled using a probability averaging technique. The final classification is performed using an SVM classifier. In this work, publicly available databases are modified (such as: DRISHTI-GS1-R, ORIGA-R, RIM-ONE2-R, LAG-R, and ACRIMA-R) based on oversampling data-level technique for validating the performance of deep ensemble model. Ensembling the best configurations reports an overall classification accuracy of 93.4%, 79.6%, 91.3%, 99.5%, and 99.6% in DRISHTI-GS1-R, ORIGA-R, RIM-ONE2-R, ACRIMA-R, and LAG-R databases, respectively.

KEYWORDS

ensemble learning, fundus image, glaucoma, pre-trained model, transfer learning

1 | INTRODUCTION

Optic nerve head, also called cranial nerve II, is a bundle of over one million nerve fibers. It carries the vision related sensory information from eyes to brain.¹ Any deterioration in the optic nerve may cause vision loss. Glaucomatous optic neuropathy (GON) is a cluster of eye disease that damages the optic nerve and causes irreversible vision loss.² As it is asymptomatic, early detection may curb vision loss.³ Furthermore, medical options are better in the early stages than the severe stage, which has a poor prognosis.⁴ Clinically, GON can be diagnosed using tonometry, perimetry, and ophthalmoscope.⁵ To aid in clinical diagnosis, various imaging modalities have been developed. Fundus imaging is the most popular among these, as it is easily affordable. However, the real-time images are captured in the presence of several factors such as illumination variations, background interference, and so forth, making GON diagnosis a challenging task. Economically, this disease imposes significant financial costs on both individuals and society, and these costs increase as the disease progresses.⁴ The cost emanates from either the requirement for advanced expertise and experience in interpreting the fundus images, or the utilization of more expensive imaging technologies.

Beyond the economic aspects, there are fundamental shortcomings of fundus photography in glaucoma screening. The visual perception of disc photographs is subjective and susceptible to interobserver variabilities. With the rising number of persons affected by glaucoma, relying on these interpretations to diagnose the disease may become untenable.⁶ To mitigate these issues, computer-aided diagnosis (CAD) based approaches are required for larger screening. CAD-based glaucoma detection from the fundus image can be broadly classified into two categories: heuristic and deep learning approaches.⁵ In heuristic methods, required features are extracted by applying the appropriate algorithm and then fed into the classifier for further classification. In some work, relevant clinical indicators are obtained from the segmented results.^{7,8} Whereas, in some approaches, the unique features are retrieved using wavelet transform techniques.^{9–11} However, these approaches rely largely on handcrafted features. This may have an impact on the effectiveness of these strategies. Furthermore, such methods may not be effective across a large range of datasets.

To address the above-mentioned limitations, deep learning (DL) algorithms could be used in glaucoma screening. The DL model learns hierarchical features from raw data and classifies them using a cascade of layers. The remarkable success of CNN has made them a promising choice for image classification task. Numerous

algorithms were developed a couple of years back for glaucoma detection using CNN. However, large amount of annotated data is required to train a deep CNN. Therefore, training process of the network has gained a lot of research attention. In general, a deep CNN can be trained in two ways: training from scratch and transfer learning with fine-tuning. Training from scratch requires large amount of labeled data, which are immensely difficult to find in medical imaging.¹² Besides, it is time-consuming and typically necessitates high computational resources. By transferring the knowledge obtained from one domain into another, transfer learning overcomes such problems. With the development of various pre-trained models, image classification using transfer learning has gained popularity.

The models can be customized for new tasks using two methods. In the first method, relevant features are extracted from the new images by utilizing pre-trained model as feature extractor. Then a new classifier is trained with these features for classification. The second technique uses the backpropagation algorithm to fine-tune the weights of the entire or part of the model. This is based on the intuition that the generic features are extracted from the earlier layers, while application-specific features are extracted from the later layers. Despite the promising outcomes, CNN suffers from high variance and may overfit due to its sensitivity to training data fluctuations. Ensemble learning helps to reduce this variance by integrating multiple models to obtain better predictive performance than could be obtained from any of the constituent models. It also improves the robustness and generalization of the model. Algorithms such as bagging, boosting, and stacking can be employed to integrate the various learners. Stacking trumps bagging and boosting in terms of enhancing accuracy.

This work makes the following notable contributions:

- i. Generation of new retinal databases from publicly available retinal databases (such as DRISHTI-GS1, ORIGA, LAG, ACRIMA, and RIM-ONE2) using oversampling data-level technique.
- ii. Implementation of several topologies of deep convolutional neural networks like Alexnet, Googlenet, VGG-16, VGG-19, Squeezenet, Resnet-18, Resnet-50, Resnet-101, Efficientnet-b0, Mobilenet-v2, Densenet-201, Inception-v3, and Xception for categorization of glaucomatous and normal fundus images.
- iii. Evaluation of pre-trained models using different classifiers like support vector machine (SVM) and k-nearest neighbor (KNN). Explore the feature reduction technique using Haar wavelet transform toward the task of classifying glaucoma and normal images.
- iv. A new ensemble framework based on single-level stacking ensemble technique is proposed for effective

classification of glaucoma and normal fundus images. This framework comprises three best CNN models, obtained through extensive experimental results search on 13 deep CNN models, which are implemented in 65 configurations. A two-stage ensemble selection technique is proposed to achieve much better results than existing state-of-the-art approaches for glaucoma classification task.

The rest of the paper is structured as follows: A brief review of glaucoma detection using pre-trained models is discussed in Section 2. In Section 3, methodology is explained. Findings and discussions are summarized in Section 4, accompanied by conclusions in Section 5.

2 | RELATED WORK

The efficacy of glaucoma detection using DL algorithms has been vastly enhanced. Prior works on automated glaucoma detection using deep learning approaches can be grouped into two categories: DL-based segmentation and DL-based classification. In the first approach, optic disc and optic cup are segmented using an appropriate DL model. With a suitable classifier, classification is done based on the computed clinical indicators. Whereas, in the second approach, vital features for classification are extracted and then classified using the DL model itself. Among the various DL models, CNN has widespread application in image classification. For glaucoma detection, either a bespoke CNN or pre-trained models can be employed. However, a brief summary of existing glaucoma detection schemes using transfer learning is discussed, as it is most useful for this research work. In some works,^{12–14} the model is trained with the region of interest (optic disc) portion of the image, as it provides the discriminative features crucial for classification. The rest of the work, on the other hand, takes into account the entire fundus image.^{4,15–21}

Fu et al.¹³ have developed a disc-aware ensemble network based on an ensemble of four independent networks. With these networks, distinct features are extracted from the fundus images. The predictions obtained by the networks are fused to achieve the final decision. With a total of 1676 (46 glaucoma and 1630 normal) and 5783 (113 glaucoma and 5670 normal) images, authors have reported an AUC of 0.918 and 0.817 in Singapore Chinese Eye Study (SCES) and Singapore Indian Eye Study (SINDI) databases, respectively. Christopher et al.¹² have demonstrated the impact of native and transfer learning of pre-trained models for glaucoma detection. By analyzing 14 822 images, the authors have

concluded that fine-tuning the Resnet-50 model outperformed the other two models, VGG-16 and Inception-v3, with an overall AUC of 0.91.

Investigation of VGG-19, Googlenet, and Resnet-50 are carried out in Reference 15. A total of 2313 fundus images obtained from RIM-ONE, DRISHTI-GS1, and ESPERANZA are used to evaluate the models. With highest AUC of 0.94, VGG-19 network surpasses the other models. Ko et al.¹⁹ have presented an ensemble approach based on VGG-16 and SVM classifier. First, the features extracted from fine-tuned VGG-16 model are utilized to compute the confidence score. Further, whenever the confidence score is lower than a threshold value, SVM classifier is used. A total of 944 fundus images from Taipei Veterans General Hospital are used. With 187 test images, they have reported an accuracy of 92.8% in the ensemble model.

Performance comparison of Inception-v3, Resnet-50, Xception, VGG-16, and VGG-19 are discussed in Reference 16. Models are assessed with 1707 images from the DRISHTI-GS1, RIM-ONE, sjchoi86-HRF, ACRIMA, and HRF databases. Authors have reported that the Xception model leads the other models with an average accuracy of 0.9605. Zhen et al.¹⁷ have examined the behavior of Densenet, Inception-v3, InceptionResnet, NasnetMobile, Resnet, VGG-16, VGG-19, and Xception for glaucoma detection. Models are trained and tested with 5978 fundus images collected from hospital. Densenet model has achieved a highest accuracy of 75.50%.

Liu et al.¹⁴ have investigated the performance of Resnet-50 for glaucoma classification. Optic disc localization is performed at the pre-processing stage. With 3768 images acquired from three ophthalmic centers and 626 images from RIM-ONE and HRF database, the efficacy of the model is examined. Sensitivity of 89.3% and specificity of 97.1% are reported in their work. Jin et al.²⁰ have used DL approach for the detection of early and advanced glaucoma. The pre-trained Inception-v3 model is fine-tuned using transfer learning approach. A total of 754 images (228 advanced, 141 early, and 385 normal) are used to train the model. Authors have reported an overall accuracy of 84.5% with 464 images (141 advanced, 87 early, and 236 normal).

Pre-trained Densenet-201 model is employed in Reference 18. The fundus images from DRISHTI-GS1 database are pre-processed for better contrast enhancement. The model's initial layers are transferred, whereas the final three layers are replaced. With 80% of images, the complete model is trained using Adam optimizer. An overall classification accuracy of 96.48% is obtained with 20% of test images. Rehman et al.²¹ have presented an ensemble framework for glaucoma detection. Predictions

obtained by four pre-trained models such as Alexnet, Inception-v3, InceptionResnet-v2, and Nasnet-Large are combined using different voting techniques. Authors have concluded that ensembling the models using accuracy based weighted voting scheme outperformed with an overall classification accuracy of 99.6%, 88.2%, 95.1%, 96.1%, and 99.8% in ACRIMA, ORIGA, RIM-ONE, Armed Forces Institute of Ophthalmology (AFIO), and Hayatabad Medical Complex (HMC) databases, respectively. An ensemble approach for the classification of glaucoma stages is presented in Cho et al.⁴ Modified versions of Inception-v3 and InceptionResnet-v2 are utilized in various strategies. Authors have obtained an improvement in the accuracy of about 3% in the ensemble method compared with single best CNN model with 3460 fundus images.

Majority of the prior research works have compared the performance of pre-trained models individually. Also, private databases are utilized in most of the existing works. It is observed that ensembling multiple models may produce better results in image classification tasks. However, limited work has been carried out toward developing an ensemble model for glaucoma diagnosis. This motivated us to develop a framework based on ensemble learning of deep models for glaucoma detection.

3 | METHODOLOGY

Ensembling multiple models minimize the variance of predictions while simultaneously producing better final predictions than any single model. Figure 1 depicts the suggested approach for glaucoma detection using the stacking ensemble technique. It includes four core blocks: generation of new retinal databases, fine-tuning of base models, ensemble selection, and classification. As deep neural networks necessitate a huge number of images, data augmentation is employed to create a larger database. Standard pre-trained models are considered as the base learners and fine-tuning these base learners is performed to obtain the predictions. A two-stage ensemble selection technique is proposed for the selection of best three base learners. The probability scores obtained by the best three base learners are aggregated. Final prediction is obtained using SVM classifier.

3.1 | Generation of retinal databases

Publicly available retinal databases like DRISHTI-GS1, RIM-ONE2, ORIGA, ACRIMA, and LAG are considered in this work. Figure 2 shows the sample glaucoma and normal images from various databases. The distribution

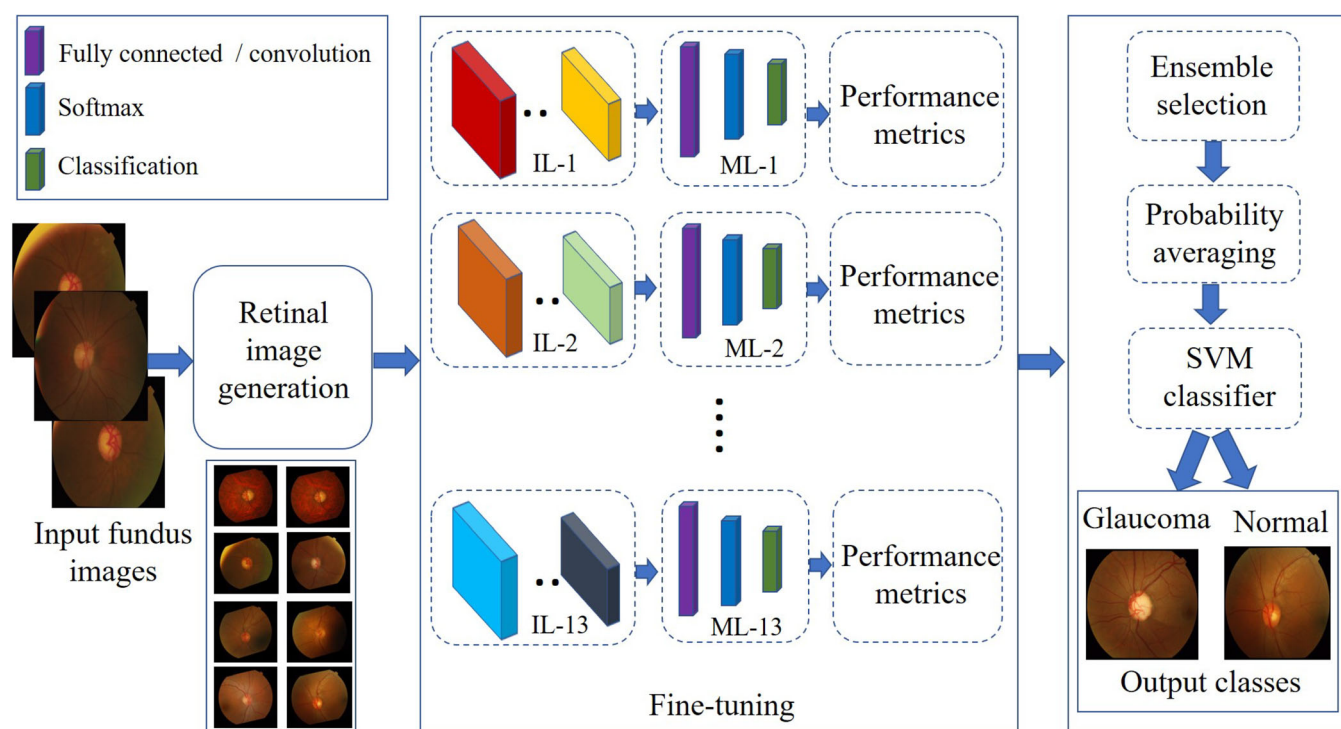


FIGURE 1 Block diagram representation of the proposed deep ensemble model for glaucoma detection from fundus images. IL-1, IL-2, ..., and IL-13 indicate the initial layers of base learner1, base learner2, ..., and base learner13. ML-1, ML-2, ..., and ML-13 represent the modified layers of the base learner1, base learner2, ..., and base learner13.

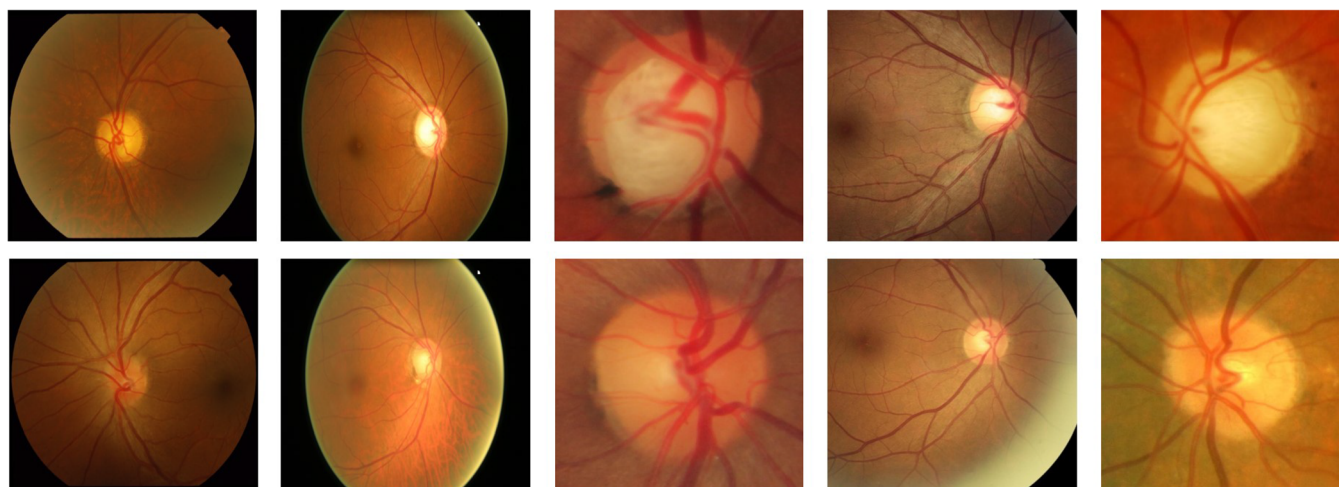


FIGURE 2 Sample glaucoma and normal images from various databases. The first and second rows indicate the sample glaucoma and normal images from DRISHTI-GS1, ORIGA, RIM-ONE2, LAG, and ACRIMA databases, respectively.

TABLE 1 Description of retinal databases

Original database	Number of images in original database	Collected from	Generated database	Number of images in generated database	Dimension and format
DRISHTI-GS1 ²²	G-89 N-12	Aravind eye hospital	DRISHTI-GS1-R	G-1691 N-1500	2896 × 1944 PNG
RIM-ONE2 ²³	G-200 N-255	Collaboration from three Spanish hospitals	RIM-ONE2-R	G-1000 N-1275	290 × 290 to 1375 × 1654 JPEG
ORIGA ²⁴	G-168 N-482	Singapore eye research center	ORIGA-R	G-1680 N-2410	3072 × 2048 JPEG
ACRIMA ¹⁶	G-396 N-309	ACRIMA project Spain	ACRIMA-R	G-3960 N-3090	178 × 178 to 1420 × 1420 JPEG
LAG ²⁵	G-1711 N-3143	Beijing Tongren hospital	LAG-R	G-3422 N-3143	500 × 500 JPEG

Note: G represents glaucoma images and N indicates normal images.

of images in the original databases is indicated in the second column of Table 1. The data distribution is skewed toward the glaucoma class in DRISHTI-GS1 and ACRIMA databases, whereas it is skewed toward the normal class in the remaining databases (RIM-ONE2, LAG, and ORIGA). Handling data imbalance problem is essential in training the DL architectures. In this work, oversampling data-level strategy is employed to handle the data imbalance problem.²⁶ The existing retinal databases are modified to enlarge the number of images using rotation augmentation technique. Except for DRISHTI-GS1, clockwise rotation with a step size of one is employed for all databases. RIM-ONE2-R database is created with 43.96% of glaucoma images and 56.04% of normal images, by rotating each image in RIM-ONE2 database between zero and four degrees. For generating ACRIMA-R database, each image is rotated between zero

and nine degrees. This results in 56.12% of glaucoma images and 43.88% of normal images. ORIGA database includes 25.84% of glaucoma images and 74.15% of normal images. Therefore, each glaucoma image is rotated between zero to nine degrees, while each normal image is rotated between zero to five degrees. This results in ORIGA-R database comprises 41.08% of glaucoma images and 58.92% of normal images. As LAG database contains 35.24% of glaucoma images, each glaucoma image is rotated between zero and one degree. Thus LAG-R database is generated with 52.12% of glaucoma images and 47.88% of normal images. DRISHTI-GS1 database includes 88.12% of glaucoma images and 11.88% of normal images. Therefore, each glaucoma image is rotated between zero and nine degrees, while each normal image is rotated between zero and 62 degrees. A step size of 0.5 is employed to generate the DRISHTI-GS1-R database,

TABLE 2 Details about pre-trained models

Pre-trained models	Number of layers	Input dimension	Transferred layers	Modified layers	Layer for feature extraction
Alexnet ²⁷	25	227 × 227 × 3	1–22	23–25	fc7: layer 20
VGG-16 ²⁸	41	224 × 224 × 3	1–38	39–41	fc7: layer 36
VGG-19 ²⁸	47	224 × 224 × 3	1–44	45–47	fc7: layer 42
Resnet-18 ²⁹	71	224 × 224 × 3	1–68	69–71	pool5: layer 68
Resnet-50 ²⁹	177	224 × 224 × 3	1–174	175–177	avg-pool: layer 174
Resnet-101 ²⁹	347	224 × 224 × 3	1–344	345–347	pool5: layer 344
Squeezenet ³⁰	68	227 × 227 × 3	1–65	66–68	fire9-concat: layer 62
Mobilenet-v2 ³¹	154	224 × 224 × 3	1–151	152–154	global-average-pooling2d-1: layer 151
Googlenet ³²	144	224 × 224 × 3	1–141	142–144	pool5-7x7-s1: layer 140
Inception-v3 ³³	315	299 × 299 × 3	1–312	313–315	avg-pool: layer 312
Xception ³⁴	170	299 × 299 × 3	1–167	168–170	avg-pool: layer 167
Densenet-201 ³⁵	708	224 × 224 × 3	1–705	706–708	avg-pool: layer 705
Efficient-b0 ³⁶	290	224 × 224 × 3	1–287	288–290	layername: layer 287

Note: layername = efficientnet-b0|model|head|global_average_pooling2d|GlobAvgPool.

which contains 52.99% of glaucoma images and 47.01% of normal images. The information regarding the original and generated retinal databases is described in Table 1.

3.2 | Fine-tuning

Transfer learning is a typical DL approach in which a model that has been trained for one task can be customized for another task. This is achieved by transferring the initial layers and optimized weights from the model. However, the final layers are modified accordingly. Pre-trained models like Alexnet, Googlenet, VGG-16, VGG-19, Squeezenet, Resnet-18, Resnet-50, Resnet-101, Efficientnet-b0, Mobilenet-v2, Densenet-201, Inception-v3, and Xception are employed in this work. Table 2 describes the specifications of various models. With more number of images, these models are trained in general. As a result, they have amassed a diverse representation of raw images. For each base model, fine-tuning is employed by unfreezing the entire model and training it on the retinal databases with a very low learning rate. It results in better performance, especially for small datasets. The model parameters are updated in order to minimize the cross-entropy loss and defined as

$$\text{loss} = - \sum_{m=1}^s \sum_{n=1}^k T_{mn} \ln(Y_{mn}), \quad (1)$$

where s is the number of samples, k is the number of classes, T_{mn} indicates the m th sample belongs to n th

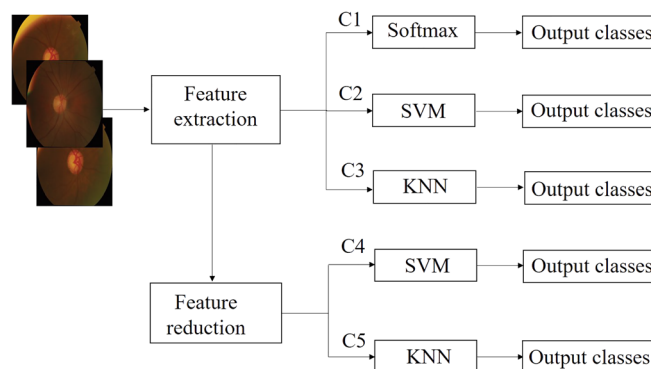


FIGURE 3 Representation of five different configurations employed for each base learner towards classifying glaucoma and normal images. With 13 CNN architectures and five various classification approaches, a total of 65 configurations are employed in this work.

class, and Y_{mn} is the likelihood that the network associates the m th input with n th class.

After training, the base learners are tested in five different configurations to obtain the performance metrics. Figure 3 shows the detail of five different configurations used in this work. For all the configurations, each base learner is completely trained using transfer learning approach. Each configuration differs in the way the final classification is performed. In Configuration 1 (C1), classification is done using the probability scores obtained by softmax layer. By incorporating softmax activation function, this layer transforms the unnormalized output of fully connected layer to a normalized output by assigning

probability scores to each class using the equation described as

$$\sigma(Y_i) = \frac{e^{Y_i}}{\sum_{j=1}^k e^{Y_j}}, \quad (2)$$

where k represents the number of classes, Y is the input vector, and $\sigma(Y_i)$ gives the output class probabilities.

Configuration 2 (C2) and Configuration 3 (C3) represent the classification obtained using SVM³⁷ and KNN³⁸ classifiers, respectively. For this, the discriminative features are extracted from the appropriate layer of the trained base learner. The objective of SVM classifier is to determine a hyperplane that maximizes the separation of the data points to their actual classes in an n -dimensional space. The main hyperparameter in SVM classifier is the kernel function. It is used to map the original dataset into a feature space, so that the observations are more easily and accurately separable after this transformation. Here, linear kernel function is employed to achieve better performance at a faster rate. The KNN classifier seeks to determine the degree of similarity between the test data and a specific number of neighboring data points in the training data. Euclidean kernel with K (number of neighbors) value of 5 is employed in KNN classifier for better classification of glaucoma and normal images. The dimension of the extracted features is reduced by applying discrete wavelet transform. In this case, the Haar wavelet³⁹ is used to perform single-level decomposition on the extracted features from the pre-trained model. The decomposition provides an approximation band and detail band. The approximation band contributes more toward glaucoma features than the detail band. Moreover, considering the approximation band greatly reduces the redundant information. This further reduces the computational burden on SVM and KNN classifier. Hence, in this work, the approximation band is used for classification by SVM and KNN in C4 and C5 configurations, respectively. For each base learner, performance metrics are obtained using five different configurations.

3.3 | Ensemble selection

After obtaining the performance metrics from 65 different configurations, ensemble selection⁴⁰ is performed for selecting the best configurations. This helps to find the best path for classification in a two-stage selection process. In the first stage, the best configurations from the five different configurations are selected for each base learner based on classification accuracy. Here 13 base learners are used, and hence 13 best configurations are

obtained in the first stage out of 65 configurations. In the second stage, the three classification approaches are selected by evaluating their classification accuracy. The outcomes of the best three learners are considered for final classification.

3.4 | Final classification

The best three predictions obtained by the base learners in the previous section are integrated using stacking ensemble technique. In this approach, the probability scores obtained by the best base learners are averaged using the following equation:

$$P_{\text{Final}}(B_k) = \left(\sum_{i=1}^N P_{s(i)}(B_k) \right) / N, \quad (3)$$

where $P_{\text{Final}}(B_k)$ represents the final probability score of B_k , k is class identifier, $P_s(B_k)$ indicates the probability score obtained from a single base learner, and N represents the number of base learners used in ensemble. Here, the values of k and N are 2 and 3, respectively. With the average probability scores, SVM classifier is used for obtaining the final classification output.

4 | RESULTS AND DISCUSSIONS

Several pre-trained models are implemented in this work for the classification of glaucoma and normal images. First, the models (base learners) that were developed primarily for the classification of multi-class problems (1000 classes) are customized to the two-class classification problem. This is accomplished by fine-tuning the models using a transfer learning approach. Then, the customized base learners are trained and tested with generated retinal databases. For each database, 70% of images are used in training phase, while the remaining unseen 30% are employed to test the base learners. Table 3 indicates the distribution of images used in the training and testing

TABLE 3 Distribution of images

Database	Number of training images	Number of testing images
DRISHTI-GS1-R	G-1178 N-1000	G-513 N-500
RIM-ONE2-R	G-700 N-895	G-300 N-380
ORIGA-R	G-1170 N-1685	G-510 N-725
LAG-R	G-2395 N-2200	G-1027 N-943
ACRIMA-R	G-2770 N-2160	G-1190 N-930

phases. In this work, the performance of each base learner is investigated in five different configurations. With batch size of 32 and maximum epochs of 20, the parameters of customized base learners are updated using Adam optimizer⁴¹ with an initial learning rate of 0.0001. Each configuration's performance is evaluated using metrics like accuracy (ACC), sensitivity (SN), specificity (SP), and precision (PR).⁵ This work is carried out on desktop with Microsoft windows 10 pro system with Intel @ Core i5-8400 CPU @ 2.80GHz, 16GB RAM. The programming environment used is MATLAB 2021a.

4.1 | First stage evaluation

A two-stage ensemble selection technique is proposed to select the top three configurations. Depending on the obtained classification accuracy, the first stage selects the best among the five configurations for each base learner. Tables 4–16 represent the obtained results for various retinal databases using different base learners. It is evident from Table 4 that the extracted features classified using softmax classifier yields a better accuracy in DRISHTI-GS1-R, RIM-ONE2-R, and ACRIMA-R. SVM classifier provides better accuracy in LAG-R and ORIGA-R with 4096 and 2048 features, respectively. For Mobilenet-v2, training and testing the SVM classifier with 1280 features provides better accuracy in RIM-ONE2-R and ACRIMA-R databases. With 640 features, the KNN classifier performs better in DRISHTI-GS1-R and ORIGA-R.

TABLE 4 Classification accuracy (ACC in %) obtained using various configurations of customized Alexnet for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R: 22	87.3	86.2	86.5	86.4	86.8
RIM-ONE2-R: 23	85.3	84.2	85.1	84.8	84.7
ORIGA-R: 24	72.8	72.7	71.6	73	71.9
ACRIMA-R: 16	97	96.8	96.4	96.2	95.8
LAG-R: 25	97.1	97.2	97	97.1	97

TABLE 5 Classification accuracy (ACC in %) obtained using various configurations of customized Mobilenet-v2 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	86	83.5	86.4	83.8	86.6
RIM-ONE2-R	84.4	85.4	84.1	84.6	84
ORIGA-R	69.8	70.2	69.6	71	71.3
ACRIMA-R	98.6	98.9	98.4	98.7	97.9
LAG-R	96.8	96.6	96.5	96.5	96.5

Configuration 1 (C1) surpasses the others in LAG-R database, with a classification accuracy of 96.8%.

From Table 6, it is clear that using 100 352 features extracted from the appropriate layer of customized Squeezenet model, the SVM classifier achieves the

TABLE 6 Classification accuracy (ACC in %) obtained using various configurations of customized Squeezenet for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	90.6	90.5	90.4	90.5	90.5
RIM-ONE2-R	81.6	82.9	80.8	82.6	80.2
ORIGA-R	74.1	71.5	73.4	71.3	73.7
ACRIMA-R	95.4	96.4	95.5	96.3	95.6
LAG-R	98.1	98.1	98	98.2	98.1

TABLE 7 Classification accuracy (ACC in %) obtained using various configurations of customized Googlenet for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	91.5	90.5	90.4	90.6	90.5
RIM-ONE2-R	81.8	81.9	82.1	81.5	81.3
ORIGA-R	76.4	75.7	75.9	76.1	76
ACRIMA-R	95.3	95.2	95.1	95.2	94.8
LAG-R	97.3	97.4	97.1	97.3	97.2

TABLE 8 Classification accuracy (ACC in %) obtained using various configurations of customized Resnet-18 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	88.5	88.6	87.6	88.5	88.4
RIM-ONE2-R	85.7	84.7	82.8	84.9	84
ORIGA-R	74.8	74.2	73.8	74.4	74.1
ACRIMA-R	99.1	98.7	98.5	98.6	98.4
LAG-R	99.2	99.1	99.2	99.1	99.3

TABLE 9 Classification accuracy (ACC in %) obtained using various configurations of customized Resnet-50 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	91.5	89.3	91.6	88.8	91.8
RIM-ONE2-R	89.9	89.3	88.1	89.7	87.8
ORIGA-R	75.6	74.2	74.7	73.4	74.5
ACRIMA-R	98.1	98.3	98.2	97.9	97.6
LAG-R	97.3	97.2	97.1	97.2	97.1

TABLE 10 Classification accuracy (ACC in %) obtained using various configurations of customized Resnet-101 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	89	83.7	84.1	83.3	83.4
RIM-ONE2-R	87.2	86.4	86.6	86.4	86.6
ORIGA-R	73.1	72.8	72.4	72.7	72.5
ACRIMA-R	99.2	98.4	98.5	98.6	99.1
LAG-R	98.8	98.7	98.5	98.7	98.6

TABLE 11 Classification accuracy (ACC in %) obtained using various configurations of customized VGG-16 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	91.5	91.7	91.9	91.7	92
RIM-ONE2-R	85.1	80.7	83.2	80.4	83.5
ORIGA-R	74.1	76.4	77.4	76.2	77.2
ACRIMA-R	96	95.5	95.3	95.7	95.4
LAG-R	95.7	97.7	97.2	97.8	97.2

TABLE 12 Classification accuracy (ACC in %) obtained using various configurations of customized VGG-19 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	90.5	90.7	90.3	90.3	90.2
RIM-ONE2-R	88.7	88.2	87.8	88.4	87.5
ORIGA-R	78.6	78.1	77.6	78.3	78.1
ACRIMA-R	95.7	97.2	96.7	97.1	96.6
LAG-R	97.2	97.6	97.2	97.7	97.2

TABLE 13 Classification accuracy (ACC in %) obtained using various configurations of customized Densenet-201 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	90.5	88	88.8	88.2	88.9
RIM-ONE2-R	90.9	90.6	90.3	90.7	90
ORIGA-R	73.6	74.3	73	74.4	72.7
ACRIMA-R	99	98.7	98.6	98.3	98.5
LAG-R	99.7	99.6	99.4	99.6	99.3

highest classification accuracy in RIM-ONE2-R and ACRIMA-R, respectively. A feature vector of 50 176 features classified using SVM classifier provides better accuracy in LAG-R database. Softmax classifier outperforms in DRISHTI-GS1-R and ORIGA-R databases. Table 7

TABLE 14 Classification accuracy (ACC in %) obtained using various configurations of customized Inception-v3 for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	91.2	90.6	90.4	90.5	90.3
RIM-ONE2-R	82.9	83.1	84.3	83.2	82.8
ORIGA-R	73.5	73.3	73.4	72.7	72.9
ACRIMA-R	99.4	98.9	99.1	98.7	98.6
LAG-R	96.7	96.5	96.9	96.6	96.8

TABLE 15 Classification accuracy (ACC in %) obtained using various configurations of customized Xception for retinal databases

Configuration/database	C1	C2	C3	C4	C5
DRISHTI-GS1-R	92.1	91.6	92	91.4	92
RIM-ONE2-R	85.3	85.2	85	85.2	85.1
ORIGA-R	72.7	71.3	72.2	71.8	72.4
ACRIMA-R	96.4	96.2	97.2	96.5	96.8
LAG-R	96	97.2	97	97.3	97.1

TABLE 16 Classification accuracy (ACC in %) obtained using various configurations of customized Efficientnet-b0 for retinal databases

Database/configuration	C1	C2	C3	C4	C5
DRISHTI-GS1-R	86.4	86	88.5	85.3	87.9
RIM-ONE2-R	86	85.4	85.8	85.3	85.6
ORIGA-R	67.4	66.3	66.2	67.1	66.7
ACRIMA-R	95.4	94.3	95.1	94.2	94.7
LAG-R	96.6	95.3	95.8	96.1	95.7

describes that features classified using softmax classifier outperform the other configurations in DRISHTI-GS1-R, ORIGA-R, and ACRIMA-R with better accuracy of 91.5%, 76.4%, and 95.3%, respectively. A feature vector of 1024 features classified using SVM and KNN classifiers yields better accuracy in LAG-R and RIM-ONE2-R, respectively. Table 8 summarizes the experimental findings of customized Resnet-18. Features classified using softmax classifier provide a better accuracy of 85.7%, 74.8%, and 99.1% in RIM-ONE2-R, ORIGA-R, and ACRIMA-R, respectively. With 512 extracted features, SVM classifier tops in DRISHTI-GS1-R, whereas KNN performs better in LAG-R with 256 features extracted from the appropriate layer of customized Resnet-18. The results obtained from different configurations of Resnet-50 is depicted in Table 9. Configuration 1 (C1) provides better results in RIM-ONE2-R, ORIGA-R, and LAG-R. With 2048

features, SVM classifier outperforms in ACRIMA-R, whereas a feature vector of 1024 features classified using KNN classifier provides better accuracy of 91.8% in DRISHTI-GS1-R database.

The classification accuracy obtained using Resnet-101 for various retinal databases in different configurations is represented in Table 10. It is inferred that softmax classifier outperforms other configurations for all databases. It is evident from Table 11 that C1 yields the highest classification accuracy of 85.1% and 96% in RIM-ONE2-R and ACRIMA-R, respectively. A feature vector of 4096 features classified using KNN classifier provides better results in ORIGA-R database. Whereas a feature vector of 2048 features classified using SVM and KNN classifiers gives better accuracy in LAG-R and DRISHTI-GS1-R databases, respectively. The obtained results for various databases using customized VGG-19 are represented in Table 12. For RIM-ONE2-R and ORIGA-R, better accuracy of 88.7% and 78.6% is achieved using softmax classifier, respectively. With 4096 features, SVM classifier outperforms the DRISHTI-GS1-R and ACRIMA-R databases. A better accuracy of 97.7% is obtained in LAG-R database, when an SVM classifier is trained and tested with extracted 2048 features from the images.

It is inferred from Table 13 that extracted features classified using softmax classifier provides better result in DRISHTI-GS1-R, RIM-ONE2-R, ACRIMA-R, and LAG-R databases. A feature vector of 960 features classified using an SVM classifier outperforms in ORIGA-R database. Table 14 indicates the metric obtained using various configurations of Inception-v3. Softmax classifier yields better results in DRISHTI-GS1-R, ORIGA-R, and ACRIMA-R. A feature vector of 1920 features classified using KNN classifier yields better accuracy in RIM-ONE2-R and LAG-R databases. Table 15 shows that better results are obtained in DRISHTI-GS1-R, RIM-ONE2-R, and ORIGA-R databases, while classifying the features using softmax classifier. In ACRIMA-R database, a feature vector of 2048 features classified using the KNN classifier produces better results. With 1024 features, SVM classifier outperforms in LAG-R databases. From Table 16, it is evident that softmax classifier outperforms in all databases, except DRISHTI-GS1-R. With 1280 features extracted from each image, KNN classifier provides better accuracy of 88.5% in DRISHTI-GS1-R database.

From the findings of first stage evaluation, it is clear that different configurations yield better results for different base learners. Furthermore, it is inferred that softmax classifier outperforms the other classifiers in 38 configurations. SVM classifier yields better results in 16 configuration, whereas KNN classifier outperforms in 11 configurations.

4.2 | Second stage evaluation

The first stage of the ensemble selection algorithm selects 13 configurations, each for an individual base learner. In the second stage, the top three among the 13 configurations are selected based on classification accuracy. Figure 4 depicts the results obtained from 13 configurations for various databases. The base learners are represented in labels (N1–N13) for better interpretation. It is inferred that, for DRISHTI-GS1-R database, Xception (N12), VGG-16 (N8), and Resnet-50 (N6) yield better accuracy of 92.1%, 92%, and 91.8%, respectively. For LAG-R database, compared with other base learners, Resnet-18 (N5), Resnet-101 (N7), and Densenet-201 (N10) provide better accuracy of 99.3%, 98.8%, and 99.7%, respectively.

Googlenet (N4), VGG-16 (N8), and VGG-19 (N9) outperform in ORIGA-R database with an overall classification accuracy of 76.4%, 77.4%, and 78.6%, respectively. All the base learners perform better in ACRIMA-R database. Specifically, Resnet-101 (N7), Densenet-201 (N10), and Inception-v3 (N11) achieves better accuracy of 99.2%, 99.1%, and 99.4%, respectively. For RIM-ONE2-R, Resnet-50 (N6), VGG-19 (N9), and Densenet-201 (N10) yield better accuracy of 89.9%, 88.7%, and 90.9%, respectively.

From the experimental findings, it is inferred that discriminative features are effectively extracted from the images using Resnet family models (Resnet-18, Resnet-50, and Resnet-101). This is due to the fact that these models have incorporated the residual connections, which greatly overcome the vanishing gradient problem in deeper networks. These models are one among the best in all databases, except ORIGA-R. The architecture of VGG-16 and VGG-19 extracts the useful features by employing a small receptive field of 3×3 in all convolutional layers. These models perform better in DRISHTI-GS1-R, RIM-ONE2-R, and ORIGA-R. Densenet-201 architecture strengthens the feature propagation using a series of dense and transition blocks. This greatly reduces the vanishing gradient problem. Due to this, Densenet-201 has become one of the best learners in LAG-R, RIM-ONE2-R, and ACRIMA-R databases. By capturing the spatial information at different scales using kernels of various sizes (such as 1×1 , 3×3 , and 5×5), Googlenet is one of the best in ORIGA-R database. Inception-v3 and Xception are the improved version of Googlenet architecture, and hence effective features can be extracted from the images. These base learners perform better in DRISHTI-GS1-R and ACRIMA-R, respectively.

The selected top three base learners are different for various databases. For each database, the probability

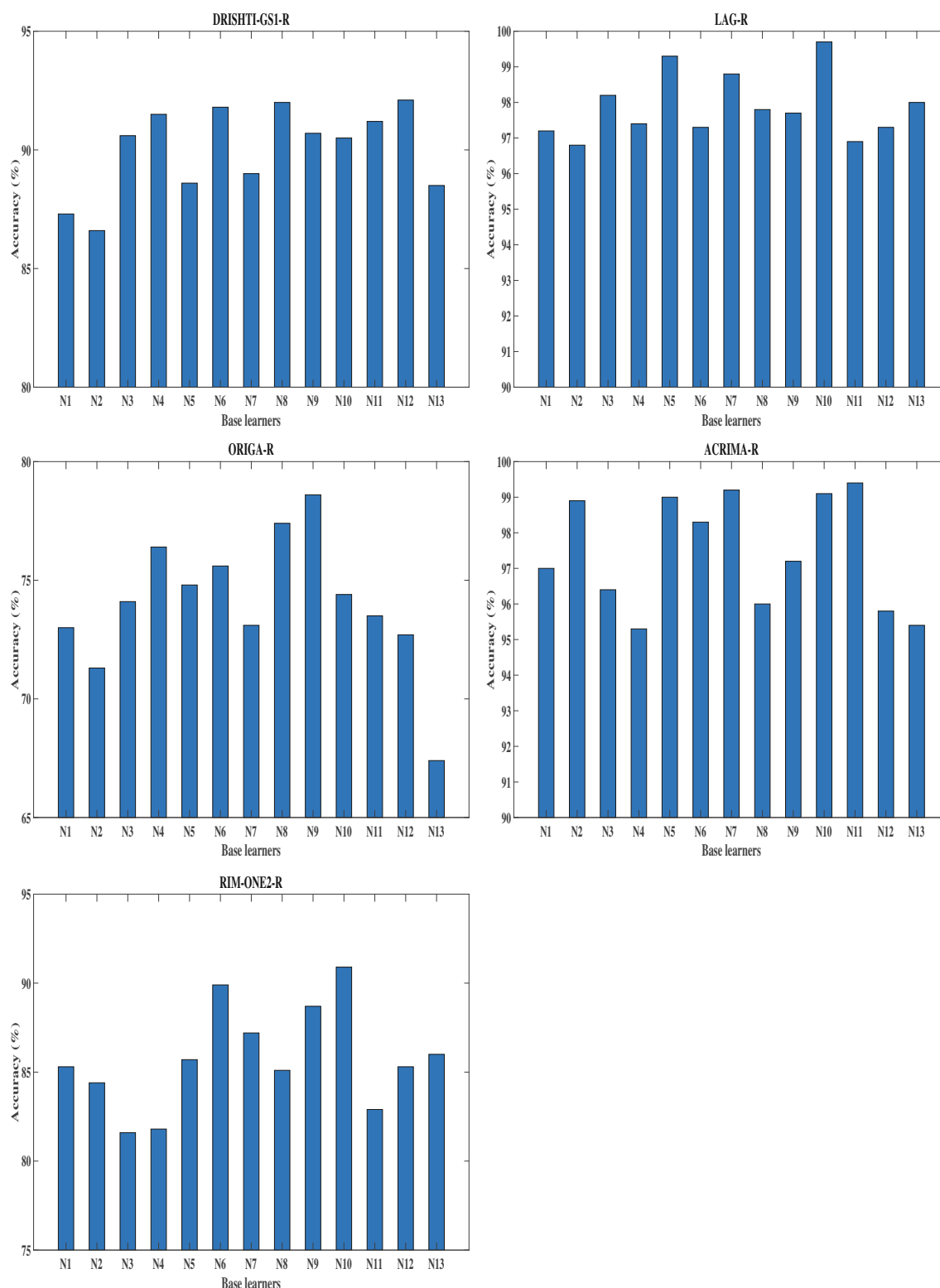


FIGURE 4 Classification accuracy obtained using various base learners for retinal databases. The base learners are represented in labels as follows: Alexnet (N1), Mobilenet-v2 (N2), Squeezenet (N3), Googlenet (N4), Resnet-18 (N5), Resnet-50 (N6), Resnet-101 (N7), VGG-16 (N8), VGG-19 (N9), Densenet-201 (N10), Inception-v3 (N11), Xception (N12), and Efficientnet-b0 (N13).

scores obtained by the best three base learners are averaged and fed into SVM classifier for final prediction. The ensemble framework achieves better classification

accuracy of 93.4%, 99.6%, 99.5%, 91.3%, and 79.6% in DRISHTI-GS1-R, LAG-R, ACRIMA-R, RIM-ONE2-R, and ORIGA-R, respectively.

TABLE 17 Performance metrics obtained by different models for retinal databases

Database and performance metrics/models	Database	ACC (%)	SN (%)	SP (%)	PR (%)
M1 [2015]	DRISHTI-GS1-R	89.4	94.7	84	85.9
M2 [2018]	DRISHTI-GS1-R	87.3	91.2	83.2	84.8
M3 [2019]	DRISHTI-GS1-R	84.1	99.4	68.4	76.7
M4 [2019]	DRISHTI-GS1-R	89.5	97.9	81	84.1
M5 [2021]	DRISHTI-GS1-R	87.1	100	73.8	79.7
M6 [2021]	DRISHTI-GS1-R	88.9	96.9	80.8	83.86
M7 [2021]	DRISHTI-GS1-R	89.4	94.7	84	85.9
Proposed	DRISHTI-GS1-R	93.4	100	86.6	88.4
M1 [2015]	ORIGA-R	78.02	55.66	93.56	85.86
M2 [2018]	ORIGA-R	71.9	50.96	86.54	72.62
M3 [2019]	ORIGA-R	71.88	61.62	79.3	69.44
M4 [2019]	ORIGA-R	75.28	60.14	85.78	74.98
M5 [2021]	ORIGA-R	70.4	52.4	83.2	68.6
M6 [2021]	ORIGA-R	78.32	58.06	92.44	84.36
M7 [2021]	ORIGA-R	78.1	64.5	89.2	81.5
Proposed	ORIGA-R	79.6	72.2	84.8	77.3
M1 [2015]	RIM-ONE2-R	80.5	75.33	84.55	78.54
M2 [2018]	RIM-ONE2-R	81.5	74.7	86.9	81.96
M3 [2019]	RIM-ONE2-R	74.08	66.84	79.74	73.1
M4 [2019]	RIM-ONE2-R	85.51	81.31	88.84	86.19
M5 [2021]	RIM-ONE2-R	81.5	78	84.2	79.6
M6 [2021]	RIM-ONE2-R	85.97	81.75	89.31	85.85
M7 [2021]	RIM-ONE2-R	87.4	88.7	86.3	83.6
Proposed	RIM-ONE2-R	91.3	91.7	91.1	89
M1 [2015]	ACRIMA-R	92.54	89.24	96.74	97.23
M2 [2018]	ACRIMA-R	93.04	91.53	95	95.88
M3 [2019]	ACRIMA-R	85.6	79.27	93.67	94.13
M4 [2019]	ACRIMA-R	96.66	95.29	93.93	88.76
M5 [2021]	ACRIMA-R	97	98.2	95.6	96.6
M6 [2021]	ACRIMA-R	96.64	96.07	97.39	97.74
M7 [2021]	ACRIMA-R	98.1	98.1	98.1	98.2
Proposed	ACRIMA-R	99.5	99.2	99.8	99.7
M1 [2015]	LAG-R	94.8	92.1	97.7	97.7
M2 [2018]	LAG-R	96.9	97.1	96.7	97
M3 [2019]	LAG-R	89.5	89.3	89.8	90.5
M4 [2019]	LAG-R	92.3	90.6	94.2	94.4
M5 [2021]	LAG-R	96.9	96.1	97.8	97.9
M6 [2021]	LAG-R	97	96.8	97.2	97.5
M7 [2021]	LAG-R	98.1	97.7	98.6	98.7
Proposed	LAG-R	99.6	99.5	99.7	99.7

Note: M1: Chen et al., M2: Raghavendra et al., M3: Juan et al., M4: Bajwa et al., M5: Ajitha et al., M6: Poonguzhali et al., M7: Rehman et al.

4.3 | Comparison with existing works

The results obtained by the proposed method are compared with other existing CNN architectures available in the literature. It is observed that the majority of existing works on CNN-based glaucoma classification employ a single CNN model. Hence for comparison, this work takes into account six CNN models developed by Chen et al.,⁴² Juan et al.,¹⁵ Bajwa et al.,⁴³ Ragavendra et al.,⁴⁴ Ajitha et al.,⁴⁵ and Poonguzhali et al.⁵ In addition, ensemble framework employed by Rehman et al.²¹ is also considered in this work. All the CNN architectures are implemented on five retinal databases. Table 17 represents the performance metrics obtained by various models in five databases. The architectures developed by Chen et al. and Bajwa et al. comprises 16 layers, whereas models developed by Ragavendra et al. and Juan et al. include a total of 20 layers. A 46-layered CNN model is developed by Ajitha et al., in which feature extraction is performed using a combination of 10 convolutional layers and 8 max pooling layers. The CNN developed by Poonguzhali et al. includes a total of 18 layers for glaucoma classification. A 16-layered compact CNN model is developed by Bajwa et al. In Rahman et al., ensemble approach using Alexnet, Inception-v3, InceptionResnet-v2, and NasnetLarge is developed.

The training hyperparameters are selected based on the information provided in the existing works. The learnable parameters are updated using SGDM optimizer in some works.^{15,21,42,43} Whereas, in some other works,^{5,44,45} Adam optimizer is used to update the parameters. With the specified hyperparameters, the models are trained and tested for various databases. It is inferred from the result that model parameters updated using Adam optimizer give better results in glaucoma classification task. From the experimental findings, it is evident that ensembling the best base learners using the stacking technique will greatly enhance the performance metrics in all five databases.

5 | CONCLUSION

A deep ensemble framework based on single-level stacking ensemble technique for the classification of glaucoma and normal images is presented in this paper. At first, 13 pre-trained models are customized for two-class classification task. Then these customized base learners are trained and tested with newly generated retinal databases like DRISHTI-GS1-R, ORIGA-R, RIM-ONE2-R, LAG-R, and ACRIMA-R in 65 configurations. Experimental result reveals that softmax classifier outperforms the other classifiers in majority of the configurations. With the proposed

two-stage ensemble selection method, the top three configurations among 13 configurations are selected. The probability scores obtained by the best three configurations are averaged. SVM classifier is trained and tested with the obtained mean probability scores. From the results, it is evident that the proposed deep ensemble model achieves better results as compared with existing state-of-the-art approaches for glaucoma classification task. Ensembling the best configurations reports an overall classification accuracy of 93.4%, 79.6%, 91.3%, 99.5%, and 99.6% in DRISHTI-GS1-R, ORIGA-R, RIM-ONE2-R, ACRIMA-R, and LAG-R databases, respectively. As a result, the suggested approach may assist ophthalmologists in diagnosing glaucoma more accurately and consistently.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author.

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